THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 12: **How Nature Works in Harmony** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In the food chain: phytoplankton, zooplankton, small fish, big fish, the secondary consumer is: 1

(a) big fish **(b)** phytoplankton **(c)** zooplankton **(d)** small fish

2. Which gas do aquatic plants release in the daytime that fish need? 1

(a) oxygen **(b)** carbon dioxide **(c)** methane **(d)** nitrogen

3. In a pond, the amount of dissolved oxygen falls when: 1

(a) it rains **(b)** plants grow
(c) it is cleaned **(d)** organic waste rots in it

4. A sudden increase in the population of rabbits in a grassland may occur if: 1

(a) owls are added **(b)** rains are heavy
(c) grass is removed **(d)** foxes are removed

5. Burning of coal increases carbon dioxide and the effect on the atmosphere is: 1

(a) cooling **(b)** more oxygen **(c)** no effect **(d)** global warming

6. A tiger population in a forest falls. The deer population will probably: 1

(a) decrease **(b)** remain zero **(c)** vanish **(d)** increase

7. If 5000 J of energy is available to producers, the energy reaching secondary consumers is about: 1

(a) 2500 J **(b)** 5 J **(c)** 50 J **(d)** 500 J

8. Which is the main cause of the rise in the CO2 level in the atmosphere? 1

(a) photosynthesis
(b) burning of fossil fuels and deforestation
(c) rainfall
(d) pollination

9. The disappearance of vultures leads to a rise in: 1

(a) cows **(b)** plants
(c) tigers **(d)** stray dogs and diseases

10. In the food web, an organism that eats both plants and animals occupies: 1

(a) more than one trophic level **(b)** no level
(c) only the first **(d)** only the third

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are there fewer top carnivores than herbivores in an ecosystem? 2

12. Why does overuse of pesticides harm an ecosystem? 2

13. Why are the plastics harmful to land and water animals? 2

14. If a grass field has 20000 J of energy, find the energy available to the tertiary consumer by the 10 per cent rule. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Construct a food chain of four organisms in a pond and identify the trophic level of each. 3

16. State how excessive hunting of wolves in a forest would affect deer and the forest vegetation. 3

17. Explain how rainwater harvesting helps in conserving water. 3

OR

Explain what happens to a food web if the population of grasshoppers decreases sharply. Use frogs and snakes. 3

18. Explain how a wetland acts as a natural water filter. 3

OR

How do forests regulate climate and rainfall? Give three points. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A grassland has grass, grasshoppers, mice, frogs, snakes and hawks. Draw the food web in words, list the chains, and predict the effect of removing the snakes. 5

OR

Compare a pond and a grassland ecosystem on the basis of producers, consumers, decomposers, abiotic factors, and give one food chain for each. 5

20. Explain the energy flow using a pyramid with 20000 J in producers. Compute each level, state the assumption and explain why chains are short. 5

OR

Explain the role of pollinators, seed dispersers and symbionts in an ecosystem with two examples for each. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND